INTRODUCTION TO R I - UMAINE

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The Jackson Laboratory

12.5 HOURS • 5 Modules • No Prior Experience Required
Part of Foundational Data Science Training Series
1-credit hour

Course Description

" How do we translate raw, messy biological data into a transparent and reproducible workflow?"

- R is a table-stakes language for data science, genomics, and bioinformatics.
- R gives you access to critical libraries for genomics, statistics, and visualization.
- R with RMarkdown/Quarto produces fully reproducible analyses — the document is the workflow.
- R skills are highly valued: see the O'Reilly Cloud Salary Survey for salary data.

This course is the first of two workshops surveying the R programming language, a central tool in bioinformatics, genomics, statistics, and data science. No prior programming experience is required. We use RStudio (Posit) throughout. By the end of this course, you will be able to write base R code confidently, manipulate and visualize data, and build the foundation for the statistical analysis covered in R II.

Course Learning Outcomes

By the end of this course, you will be able to:

- Navigate the Posit (RStudio) interface and use Quarto/RMarkdown to document analysis reproducibly.
- Understand and apply R's core data structures: vectors, lists, data frames, and matrices.
- Write base R code for importing, transforming, and summarizing data.
- Define custom functions and use apply-family functions to avoid repetitive code.
- Visualize data appropriately using base R plots: histograms, boxplots, scatterplots, mosaic plots.
- Compute descriptive statistics and recognize how confounding variables can reverse conclusions.
- Confident writing base R code, manipulating data frames, and producing and interpreting a suite of base R visualizations and descriptive statistics.

Course Overview

Introduction to R I and Introduction to R II form a complete, two-course sequence. Together they take you from zero programming experience to conducting, visualizing, and statistically interpreting biological data analyses in a fully reproducible workflow.

Introduction to R I

"Learn R as a programming language"

- Base R syntax & data types
- Data structures: vectors, lists, data frames, matrices
- Control flow, loops, custom functions
- Data wrangling & cleaning
- Base R visualization & descriptive statistics

Introduction to R II

"Use R as a data-analysis platform"

- Tidyverse philosophy & dplyr
- ggplot2 for publication-quality visualization
- Hypothesis testing (Chi-squared, Fisher's exact, t-tests)
- ANOVA & nonparametric alternatives
- Correlation & linear regression

Pre-requisites:

- RStudio and the R Programming language must **both** be installed **prior** to the course start date so we are ready to go! They are both available for free, but – depending on your laptop permissions, you might need to ask IT for installation help.
  - For JAX: You can download both of them from the JAX software catalogue on SharePoint
  - For non-JAX: You can download R from https://www.r-project.org/ and RStudio from Posit

Undergraduate Credit Track

ABOUT

This section applies only to students enrolled for university credit through the University of Maine. Internal-only JAX participants follow the standard track and do not submit the items listed here.

Credit Requirements

1 credit unit (~40 hours total: 12.5 hours contact time + independent project work)

- Attend all five modules and actively participate in in-class exercises.
- Complete all five standard module assignments (same as internal students).
- Complete the five undergraduate credit questions (one per module) using the JAXMouse dataset.
- Submit each module's Project Milestone file by the date posted on Canvas.
- Your Module 5 compiled report forms the R I component of the semester project, continued in R II.

Grade Rubric

This course is graded on a **100-point scale** broken down as follows:

1. Five short assignments, each assignment is worth 7 points. These assignments are large completed collaboratively during class. **Total = 35 points**
2. Five-part project, each part is worth 13 points. **Total = 65 points**

Grading Philosophy: Because this is a skills-based course, grading emphasizes completion, demonstrated effort with commenting your code and in Markdown blocks, and functional code rather than differences in coding proficiency or efficiency.

Shared Dataset: JAXMouse Gene Expression (Simulated)

All UMaine credit questions across both courses use a single simulated RNA-seq-style dataset: JAXMouse_GeneExpression_Simulated.csv, available on the course GitHub site.

Column	Description
SampleID	Unique sample identifier (e.g., Brain_CTL_Rep1)
Tissue	Factor: Brain, Liver, Muscle
Treatment	Factor: Control, Treated
Replicate	Integer: 1, 2, or 3
Sex	Factor: Male, Female
Age_weeks	Numeric: mouse age at sacrifice (weeks)
Gene	Character: gene symbol (e.g., Actb, Gapdh, Brca1)
Expression	Numeric: log2 CPM expression value
QC_pass	Logical: TRUE / FALSE quality control flag
BatchID	Factor: Batch1, Batch2 (used to illustrate Simpson's Paradox)

Project Milestones: One Per Module

Module	Milestone Deliverable	Due
Module 1	JAXMouse_clean.csv (factors converted, columns verified)	*See GitHub*
Module 2	JAXMouse_gene_summary.csv (mean, SD, outlier flag per gene)	*See GitHub*
Module 3	JAXMouse_QCpass.csv (final clean, QC-filtered dataset)	*See GitHub*
Module 4	JAXMouse_baseR_plots.pdf (base R visualization suite)	*See GitHub*
Module 5	JAXMouse_RI_Analysis report (compiled Quarto HTML)	*See GitHub*

Schedule & Content

Module	Topic & Content	Resources
	MODULE 0: Setup & Orientation *Q: How do we ensure we are ready to hit the ground running on Day 1?* *A: Install R and RStudio (Posit) before the first session and verify that both run correctly.*	
0	**Setting Up Your Environment** **Learning Objectives** • Download and install R (the programming language) from CRAN. • Download and install RStudio / Posit (the graphical interface). • Launch RStudio and confirm that the Console, Environment, and Files panes are visible. • Run a simple command (e.g., 1 + 1) to confirm R is working. **Activities** • Use the JAX Software Catalogue or the Self-Service icon to install R and RStudio OR go to the R and RStudio websites • Contact IT if installation is blocked —> do this before Day 1.	**Installation** • JAX Software Catalogue (internal) • Posit/RStudio (External website) • R Programming Language (External website) **General R Reference** • R for Data Science (Hadley Wickham) • Intro to R (online book) **Statistics Reference** Stats Flowchart
	MODULE 1: Introduction to R, RStudio & Reproducible Reporting *Q: How do we ensure our data analysis is fully transparent and reproducible, for others and for our future selves?* *A: We move away from ad-hoc spreadsheets to RMarkdown/Quarto, where the document itself encodes the entire analytical workflow.*	
1AB	**Learning Objectives** • Navigate the four panes of the RStudio/Posit interface (Console, Script Editor, Environment, Files). • Create a new .qmd (Quarto) file and compare it to the legacy .rmd format. • Write and render Markdown text, including headers, lists, and inline formatting. • Insert and execute R code chunks within a Quarto document. • Knit/render the document to HTML or Word. • Import an external .csv file into the R environment and perform a first inspection of the data frame. **In-Class Activities** • Work through Module_1A notebook together. • Work through Module_1B notebook.	**Resources:** • R Markdown Cheat Sheet **Assignment for Certificate of Completion:** See GitHub Course Site.

- Import a provided data frame and explore it using head(), str(), and summary().
- Create, save, and knit a .qmd file as the first assignment.

MODULE 2: Data Types & Structures in R

Q: How does the "shape" and type of our data dictate the code we must write to analyze it?

A: Different data structures (vectors, factors, matrices, lists, data frames) each have rules, understanding them is the key to writing correct, efficient R code.

2AB

Learning Objectives

- Distinguish between R's core data types: numeric, integer, character, logical, factor.
- Create and manipulate vectors, factors, matrices, lists, and data frames.
- Understand why factors are critical for categorical biological data.
- Read and write .csv files; handle missing values with na.omit() and na.rm.
- Use subset() and tapply() to interrogate data frames.
- Explain what coercion is and when it causes unexpected results.

In-Class Activities

- Work through Module_2AB notebook.
- Compare and contrast vectors, data frames, and different data types using example biological data.

Key functions

- read.csv(), na.omit(), factor() vs. numeric
- Load a .csv from GitHub and summarize

Assignment for Certificate of Completion: See GitHub Course Site.

MODULE 3: Programming in R: Loops, Functions & Apply

Q: When and how should we generalize a specific solution into a reusable function?

A: We move beyond single commands and think in terms of bundling iterative processes into loops, apply functions, and custom functions.

3AB

Learning Objectives

- Identify and use built-in R functions with required, optional, and default arguments.
- Write custom named functions and understand the syntax of anonymous functions.
- Use for loops and understand when iteration is appropriate.
- Apply higher-order functions — sapply(), lapply(), Map(), Filter() — to avoid repetitive code.
- Explain the difference between a function's inputs, body, and return value.
- (Optional) Explore the swirl() interactive package for extra practice.

In-Class Activities

- Work through Module_3AB notebook.
- Write small functions that clean and summarize biological data.
- Build a basic calculator to practice function syntax.

Apply family

apply() family of functions (tutorial)

Assignment for Certificate of Completion: See GitHub Course Site.

MODULE 4: Data Wrangling, Cleaning & Exploratory Visualization

Q: How do we tame messy real-world data, and what assumptions and relationships are hidden in our visualizations?

*A: We learn to clean and reshape raw datasets using slicing, Boolean logic, and the *apply family, then use base R plots to reveal structure, uncertainty, and conditional relationships.*

4A

Learning Objectives

- Slice data frames using indexing, logical / Boolean operators, which(), and subset().
- Handle missing data systematically and document decisions.
- Use tapply() with multiple grouping factors to produce summary tables.
- Visualize uncertainty using boxplots (including grouped boxplots).
- Choose the appropriate plot type for the data: scatterplot, histogram, boxplot, mosaic plot.
- Read and interpret mosaic plots as conditional probability displays.
- Explain what assumptions are embedded in common visualizations.

In-Class Activities

- Work through Module_4A and 4B (visualization) notebooks.
- Clean and format a messy dataset for analysis.
- Explore a new dataset and produce a suite of base R visualizations.

Visualization guides

- From Data to Viz
- R Cookbook — Graphs
- Tidy Data Tutor
- R4DS — Data Viz chapter

Assignment for Certificate of Completion: See GitHub Course Site.

MODULE 5: Descriptive Statistics, Simpson's Paradox & Course Synthesis

Q: How can basic exploratory statistics both reveal the truth and trap us in statistical paradoxes?

A: We use the biostats package to compute descriptive statistics on simulated clinical data and discover how ignoring a confounding variable can completely reverse our conclusions

5AB

Learning Objectives

- Use the biostats package to simulate clinical datasets and compute descriptive statistics: mean, median, SD.
- Identify and report missing values systematically.
- Explain Simpson's Paradox and recognize the conditions that produce it.
- Synthesize the week's skills: from data import and cleaning through to visualization and summary statistics.
- Formulate a testable null hypothesis — the bridge to formal testing in R II.

In-Class Activities

- Work through Module_5AB
- Simulate a Drug A vs. Placebo clinical trial dataset using biostats.
- Demonstrate Simpson's Paradox with the simulated data.
- Closing Q&A: review any questions from the week.

biostats package
biostats README

Assignment for Certificate of Completion: See GitHub Course Site.

Further Learning & Enrichment Resources

The resources below supplement the course content. None are required; they are provided for students who want to go deeper.

Resource	URL / Notes
R for Data Science (R4DS)	https://r4ds.hadley.nz/
Intro to R (online book)	https://intro2r.com/
R Markdown Cheat Sheet	https://www.rstudio.com/wp-content/uploads/2015/02/rmarkdown-cheatsheet.pdf
From Data to Viz	https://www.data-to-viz.com
R Cookbook — Graphs	http://www.cookbook-r.com/Graphs/
Statistics flowchart	https://www.statsflowchart.co.uk
apply() family tutorial	https://ademos.people.uic.edu/Chapter4.html
biostats package README	https://cran.r-project.org/web/packages/biostats/readme/README.html